Practice workbook · Area of a triangle using half ab sin C

Using the area formula (1/2)ab sin C for a triangle, forwards or backwards, including the case where a given area allows an acute or an obtuse included angle.

12 question parts: 1 worked-example retry and 11 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

OLevels.com SCHOOL

Type ID: FPM-TRI-11

1. Question 1 · part whole · 4 marks

2024 JUN · Paper 01

Worked-example retry · Attempt part whole only.

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

1 In triangle ABC, $AB = 2x$ cm, $BC = 3x$ cm and $AC = 4x$ cm

The area of triangle ABC is 50 cm^2

Find, to 2 decimal places, the value of x

(4)

OLevels.com — SCHOOL —

(Total for Question 1 is 4 marks)

1. Question 1 · part whole · 4 marks · Working and answer

Working and answer

OLevels.com SCHOOL

2. Question 2 · part b · 2 marks

2019 JUN · Paper 01 (regional R)

Further practice · Attempt part b only.

2

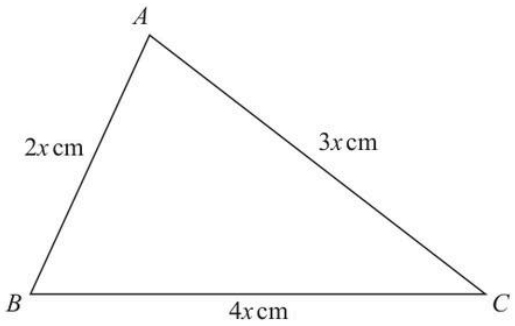


Diagram **NOT** accurately drawn

Figure 2

Figure 2 shows triangle ABC in which

$AB = 2x$ cm $\quad AC = 3x$ cm $\quad BC = 4x$ cm

(a) Show that $\sin ABC = \frac{3\sqrt{15}}{16}$

(4)

Given that the area of triangle ABC is $\frac{75\sqrt{15}}{64}$ cm^2

(b) find the value of x.

(2)

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

OLevels.com SCHOOL

4

2. Question 2 · part b · 2 marks

2019 JUN · Paper 01 (regional R)

Further practice · Attempt part b only.

Question 2 continued

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

OLevels.com SCHOOL

(Total for Question 2 is 6 marks)

2. Question 2 · part b · 2 marks · Working and answer

Working and answer

OLevels.com SCHOOL

3. Question 2 · part b · 3 marks

2023 NOV · Paper 02

Further practice · Attempt part b only.

2 In triangle ABC, $AB = 3x$ cm, $BC = 5x$ cm and $\angle ABC = 110°$

(a) Find, in degrees to one decimal place, the size of $\angle BCA$

(4)

The area of triangle ABC is 24 cm^2

(b) Find, to 3 significant figures, the value of x

(3)

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

OLevels.com SCHOOL

3. Question 2 · part b · 3 marks

2023 NOV · Paper 02

Further practice · Attempt part b only.

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

Question 2 continued

OLevels.com SCHOOL

(Total for Question 2 is 7 marks)

3. Question 2 · part b · 3 marks · Working and answer

Working and answer

OLevels.com SCHOOL

4. Question 3 · part b · 4 marks

2020 NOV · Paper 02 (regional R)

Further practice · Attempt part b only.

3

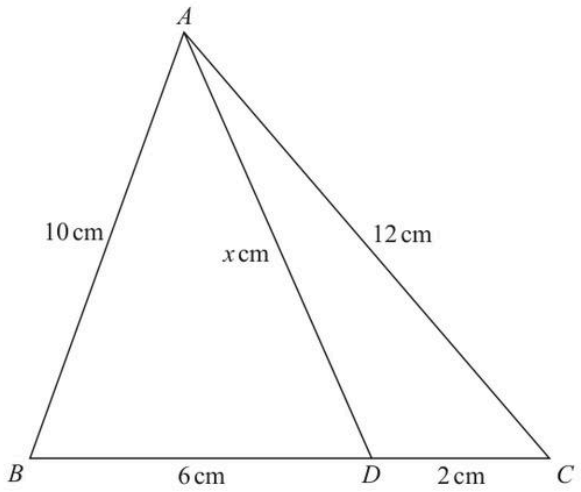


Diagram **NOT** accurately drawn

Figure 1

Figure 1 shows the triangle ABC in which $AB = 10\,\text{cm}$ and $AC = 12\,\text{cm}$.
The point D lies on BC such that $BD = 6\,\text{cm}$, $DC = 2\,\text{cm}$ and $AD = x\,\text{cm}$.

(a) Show that $x = 11$

(4)

(b) Find the area, in cm^2 to 3 significant figures, of triangle ADB.

(4)

DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA

P 6 2 2 8 4 A 0 6 3 2

4. Question 3 · part b · 4 marks

2020 NOV · Paper 02 (regional R)

Further practice · Attempt part b only.



Question 3 continued

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

OLevels.COM — SCHOOL —

(Total for Question 3 is 8 marks)

4. Question 3 · part b · 4 marks · Working and answer

Working and answer

OLevels.com SCHOOL

5. Question 3 · part a · 3 marks

2024 JUN · Paper 01 (regional R)

Further practice · Attempt part a only.

3

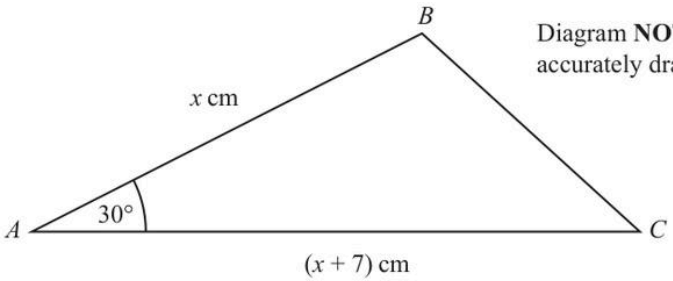


Figure 1

Figure 1 shows triangle ABC where

$AB = x$ cm $\quad AC = (x + 7)$ cm $\quad \angle BAC = 30°$

The area of triangle $ABC = 36$ cm^2

(a) Show that $x = 9$ **(3)**

(b) Find, in cm to 3 significant figures, the length of BC **(2)**

(c) Find, in degrees to one decimal place, the size of

(i) $\angle ABC$

(ii) $\angle ACB$ **(3)**

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

OLevels.COM — SCHOOL —

P 7 4 0 9 9 A 0 6 3 2

5. Question 3 · part a · 3 marks · Working and answer

Working and answer

OLevels.com SCHOOL

6. Question 5 · part b · 3 marks

2025 JUN · Paper 02

Further practice · Attempt part b only.

5 In triangle ABC, $AC = 12$ cm, $BC = 14$ cm, $AB = x$ cm and angle $ABC = 30°$

(a) Show that $x = P\sqrt{3} \pm \sqrt{95}$ where P is a prime number.

(5)

(b) Hence or otherwise, find in cm^2, the difference between the two possible areas of triangle ABC

Give your answer in the form $m\sqrt{n}$ where m is a prime number and n is an integer.

(3)

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

OLevels.com SCHOOL

6. Question 5 · part b · 3 marks

2025 JUN · Paper 02

Further practice · Attempt part b only.

Question 5 continued

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

OLevels.COM — SCHOOL —

(Total for Question 5 is 8 marks)

6. Question 5 · part b · 3 marks · Working and answer

Working and answer

OLevels.com SCHOOL

7. Question 6 · part c · 3 marks

2022 JUN · Paper 01 (regional R)

Further practice · Attempt part c only.

6 Given that $\dfrac{a+\sqrt{5}}{\sqrt{5}-2} = 11 + 5\sqrt{5}$

(a) without using a calculator, find the value of a
Show your working clearly.

(2)

Triangle PQR is such that

$PR = (x+3)\,\text{cm}$ $QR = x\,\text{cm}$ angle $QPR = 30°$ angle $PQR = 45°$

(b) Show that $x = 3 + 3\sqrt{2}$

(3)

Given that $\sin 105° = \dfrac{\sqrt{6}+\sqrt{2}}{4}$ and that the area of triangle PQR is $A\,\text{cm}^2$

(c) find the exact value of A in the form $\dfrac{9}{8}\left(p\sqrt{6} + q\sqrt{2} + r\sqrt{3} + s\right)$

where p, q, r and s are integers.

(3)

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

OLevels.com SCHOOL

P 7 1 6 4 1 A 0 1 2 3 6

7. Question 6 · part c · 3 marks

2022 JUN · Paper 01 (regional R)

Further practice · Attempt part c only.

Question 6 continued

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

OLevels.com SCHOOL

P 7 1 6 4 1 A 0 1 3 3 6

13

Turn over

7. Question 6 · part c · 3 marks

2022 JUN · Paper 01 (regional R)

Further practice · Attempt part c only.

Question 6 continued

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

OLevels.com SCHOOL

7. Question 6 · part c · 3 marks

2022 JUN · Paper 01 (regional R)

Further practice · Attempt part c only.

Question 6 continued

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

OLevels.com SCHOOL

(Total for Question 6 is 8 marks)

7. Question 6 · part c · 3 marks · Working and answer

Working and answer

OLevels.com SCHOOL

8. Question 3 · part c · 4 marks

2023 JUN · Paper 01 (regional R)

Further practice · Attempt part c only.

3

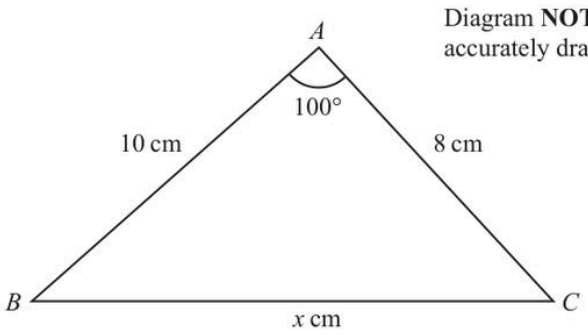


Figure 2

Figure 2 shows triangle ABC where

$$AB = 10 \text{ cm}, \quad AC = 8 \text{ cm}, \quad BC = x \text{ cm} \quad \text{and} \quad \angle BAC = 100^\circ$$

(a) Find, to 3 significant figures, the value of x

(2)

(b) Find, in degrees to one decimal place, the size of

(i) angle ABC (ii) angle ACB

(3)

The bisector of angle ABC meets AC at the point M

(c) Find the area, in cm^2 to 3 significant figures, of triangle BMC.

(4)

DO NOT WRITE IN THIS AREA DO NOT WRITE IN THIS AREA DO NOT WRITE IN THIS AREA

P 7 4 0 9 8 A 0 6 3 2

8. Question 3 · part c · 4 marks

2023 JUN · Paper 01 (regional R)

Further practice · Attempt part c only.

Question 3 continued

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

OLevels.com SCHOOL

(Total for Question 3 is 9 marks)

8. Question 3 · part c · 4 marks · Working and answer

Working and answer

OLevels.com SCHOOL

9. Question 8 · part b · 4 marks

2024 NOV · Paper 02

Further practice · Attempt part b only.

8

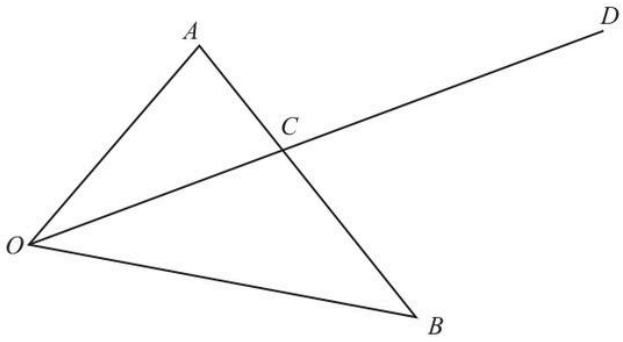


Diagram **NOT** accurately drawn

Figure 2

Figure 2 shows triangle AOB

$\overrightarrow{OA} = 4\mathbf{a} + 5\mathbf{b}$ $\overrightarrow{OB} = 8\mathbf{a} - \mathbf{b}$ $\overrightarrow{OD} = 15\mathbf{a} + 10\mathbf{b}$ where $|\mathbf{a}| = |\mathbf{b}| = 1$

(a) (i) Find $\overrightarrow{AB}$ in terms of **a** and **b**

(ii) Find, in its simplest form, the exact value of $|\overrightarrow{AB}|$

(3)

(b) Find the area of triangle AOB

(4)

The point C lies on AB and OD such that O, C and D are collinear.

(c) Use a vector method to find vector $\overrightarrow{OC}$ as a simplified expression in terms of **a** and **b**

(5)

DO NOT WRITE IN THIS AREA DO NOT WRITE IN THIS AREA DO NOT WRITE IN THIS AREA

20

P 7 3 5 8 7 A 0 2 0 3 2

9. Question 8 · part b · 4 marks · Working and answer

Working and answer

OLevels.com SCHOOL

10. Question 3 · part b · 3 marks

2025 NOV · Paper 01

Further practice · Attempt part b only.

3

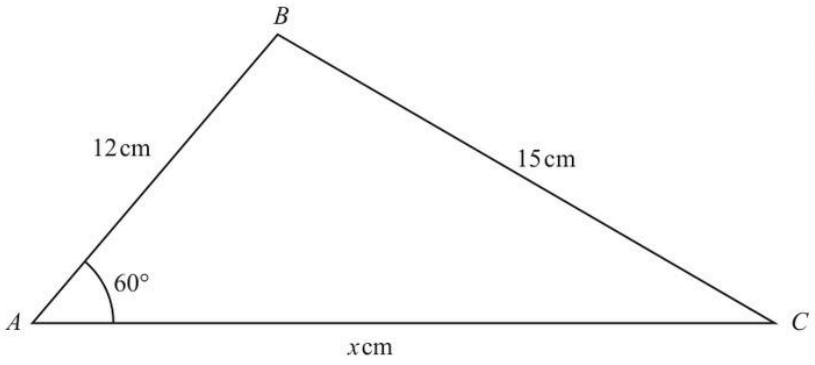


Diagram **NOT** accurately drawn

Figure 1

Figure 1 shows triangle ABC where

$AB = 12$ cm $\quad BC = 15$ cm $\quad AC = x$ cm $\quad \angle BAC = 60°$

(a) Show that x satisfies the equation

$$x^2 - 12x - 81 = 0$$

(3)

(b) Hence find the exact area, in cm^2, of triangle ABC

Give your answer in the form $9\left(p\sqrt{q} + \sqrt{r}\right)$ where p, q and r are integers.

(3)

DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA

P 7 8 9 6 4 A 0 6 3 6

10. Question 3 · part b · 3 marks

2025 NOV · Paper 01

Further practice · Attempt part b only.

Question 3 continued

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

OLevels.com SCHOOL

(Total for Question 3 is 6 marks)

10. Question 3 · part b · 3 marks · Working and answer

Working and answer

OLevels.com SCHOOL

11. Question 3 · part whole · 5 marks

2020 JAN · Paper 02 (regional R)

Further practice · Attempt part whole only.

3 In triangle ABC, $AB = 11\,\text{cm}$ and $BC = 12\,\text{cm}$.

The area of triangle $ABC = 33\,\text{cm}^2$

Find, in cm to 3 significant figures, the two possible lengths of AC.

(5)

DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA

OLevels.com SCHOOL

11. Question 3 · part whole · 5 marks

2020 JAN · Paper 02 (regional R)

Further practice · Attempt part whole only.

Question 3 continued

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

OLevels.com SCHOOL

(Total for Question 3 is 5 marks)

11. Question 3 · part whole · 5 marks · Working and answer

Working and answer

OLevels.com SCHOOL

12. Question 7 · part b · 7 marks

2021 NOV · Paper 01

Further practice · Attempt part b only.

7

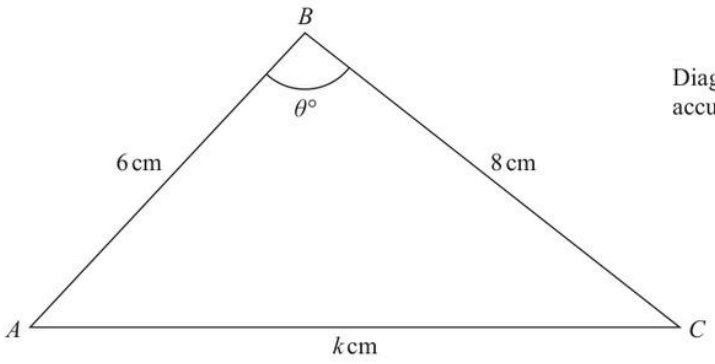


Diagram **NOT** accurately drawn

Figure 2

Figure 2 shows triangle ABC

$AB = 6\,\text{cm}$ $BC = 8\,\text{cm}$ $AC = k\,\text{cm}$ $\angle ABC = \theta°$

(a) Show that $\cos\theta° = \dfrac{100 - k^2}{96}$ (2)

The area of triangle ABC is $\sqrt{455}$ cm^2

(b) Find the two possible values of k (7)

OLevels.com SCHOOL

DO NOT WRITE IN THIS AREA DO NOT WRITE IN THIS AREA DO NOT WRITE IN THIS AREA

12. Question 7 · part b · 7 marks

2021 NOV · Paper 01

Further practice · Attempt part b only.

Question 7 continued

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

OLevels.com SCHOOL

12. Question 7 · part b · 7 marks

2021 NOV · Paper 01

Further practice · Attempt part b only.

Question 7 continued

DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA

OLevels.com SCHOOL

16

P 6 6 0 2 6 R A 0 1 6 3 2

12. Question 7 · part b · 7 marks

2021 NOV · Paper 01

Further practice · Attempt part b only.

Question 7 continued

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

OLevels.COM SCHOOL

(Total for Question 7 is 9 marks)

12. Question 7 · part b · 7 marks · Working and answer

Working and answer

OLevels.com SCHOOL